Daily RO Production Dashboard

Quick Start Guide — v2.0

RO Writer (Microsoft SQL Server) Ready

Overview

This dashboard delivers real-time visibility into Repair Order labor hours, productivity patterns, efficiency, and weekly performance. It supports both **Microsoft SQL Server** (RO Writer) and demo mode with realistic mock data.

Key Features: Daily KPIs • Hours per RO • Hourly Monitoring • 7/30-day Trends • Efficiency Tracking • Weekly Production Tracker (Excel-style)

System Requirements

Component	Requirement	Notes
Operating System	Windows, Linux, or macOS	Windows recommended for MSSQL
Python	3.10 or higher	Required for Streamlit version
Web Browser	Chrome, Edge, or Firefox	Modern browser required
Database	Microsoft SQL Server	RO Writer database
Network	Access to SQL Server	Firewall rules may apply

Dependencies

Python Packages (Streamlit version)

```
pip install streamlit pandas numpy plotly openpyxl requests pyodbc python-dotenv
```

Microsoft SQL Server Requirements

• **pyodbc** Python package
• **Microsoft ODBC Driver 17 or 18** for SQL Server
- Windows: Usually pre-installed
- Linux/macOS: Install from Microsoft

HTML Frontend

No installation required — single HTML file that works in any modern browser.

Installation Steps

1. Download the Project

Download and unzip the project folder containing:
• **ro_production_dashboard/** (Streamlit app + docs)
• **ro_dashboard_frontend/** (HTML frontend)

2. Install Python Dependencies

```
cd ro_production_dashboard
pip install -r requirements.txt
```

3. Install ODBC Driver (MSSQL)

```
Windows: Usually already installed.
Linux (Ubuntu/Debian):
curl https://packages.microsoft.com/keys/microsoft.asc | sudo apt-key add -
curl https://packages.microsoft.com/config/ubuntu/$(lsb_release -rs)/prod.list | sudo tee
/etc/apt/sources.list.d/mssql-release.list
sudo apt-get update
sudo ACCEPT_EULA=Y apt-get install -y msodbcsql17
```

Configuration

Option A: Mock Data (Recommended for Testing)

No configuration needed. The dashboard runs with realistic sample data out of the box.

Option B: Connect to Real RO Writer Database (MSSQL)

1. Create **.streamlit/secrets.toml** inside the **ro_production_dashboard** folder:

```
[mssql] connection_string = "DRIVER={ODBC Driver 17 for SQL Server};SERVER=your-server;DATABASE=YourRO
WriterDB;UID=readonly_user;PWD=YourPassword;TrustServerCertificate=yes"
```

2. In **app.py**, set: **USE_MOCK_DEFAULT = False**
3. Customize the SQL query inside **fetch_data()** to match your RO Writer table structure.

Running the Dashboard

Streamlit Version (Recommended for MSSQL)

```
cd ro_production_dashboard
streamlit run app.py
```

Opens at **http://localhost:8501**. Toggle mock mode and click **Generate Report**.

HTML Frontend (Quick Demo)

Simply open **ro_dashboard_frontend/index.html** in any modern browser.
Click **Generate Report** and use the **View Weekly** button for the Excel-style tracker.

Main Features

Feature	Location	Description
Daily Overview	Main Tab	KPIs + sortable Hours per RO table
Hourly Monitoring	Tab 2	Bar chart + day-over-day line comparison
Trends	Tab 3	7-day and 30-day charts (hours + efficiency)
Day Comparison	Tab 4	Side-by-side metrics with variance
Weekly Tracker	Sidebar Button	Excel-style view with Earned/Goal per day

Efficiency Formula: Actual Hours ÷ (Unique Technicians × 7 expected hours per day)

Troubleshooting

Problem	Solution
pyodbc not found	Run: pip install pyodbc
MSSQL connection error	Check connection string and firewall
Wrong columns in results	Edit the SQL query in fetch_data()
Charts look broken	Use a modern browser (Chrome/Edge)

For advanced configuration and full documentation, see the **README.md** file included with the project.

— End of Quick Start Guide —